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arXiv:2605.06604 (q-fin)

[Submitted on 7 May 2026]

A Geometry-Aware Residual Correction of Hagan's SABR Implied Volatility Formula

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This paper proposes a hybrid methodology to improve the approximation of SABR (Stochastic Alpha Beta Rho) implied volatility by combining analytical structure with machine learning. The approach augments the neural-network input representation with geometric features derived from the stochastic differential equations of the SABR model. Unlike approaches that fully replace analytical formulas with black-box models, the proposed framework preserves the analytical backbone of the model. The hybridization operates along two complementary dimensions. First, geometry-aware variables reflecting intrinsic properties of the SABR dynamics are used as structured inputs to the network. Second, the neural network is trained to learn the residual error relative to Hagan's closed-form approximation rather than implied volatility directly. The resulting model acts as a structured residual correction to the

analytical formula, retaining interpretability while capturing higher-order effects that are not included in the asymptotic expansion. Numerical experiments conducted over realistic parameter domains, as well as stressed environments, show that the method improves accuracy and robustness compared with both analytical approximations and standard neural-network approaches. Because the correction remains lightweight and structurally consistent with the underlying model, the framework is well suited for real-time pricing and calibration in practical trading environments.

Comments: 33 pages, 17 figures

Subjects: **Computational Finance (q-fin.CP)**; Machine Learning (stat.ML)

Cite as: [arXiv:2605.06604](https://arxiv.org/abs/2605.06604) [q-fin.CP]

(or [arXiv:2605.06604v1](https://arxiv.org/abs/2605.06604v1) [q-fin.CP] for this version)

<https://doi.org/10.48550/arXiv.2605.06604>

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Submission history

From: Adil Reghai [[view email](#)]

[v1] Thu, 7 May 2026 17:22:39 UTC (6,086 KB)

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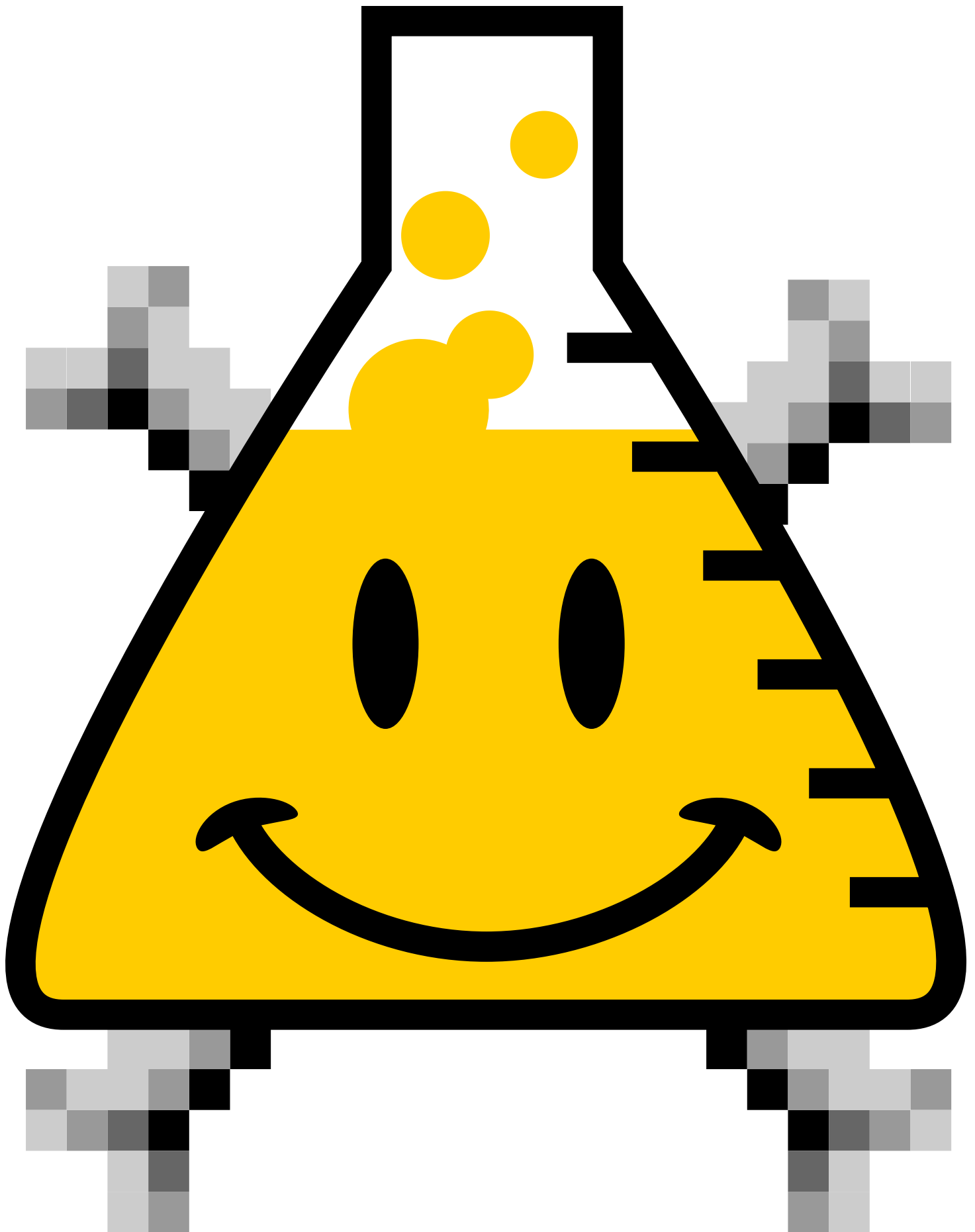
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